

Geometry supervision for 3DGRUT

Depth and normal quality on OB3D: what worked, what did not, and why

Ablation report — all numbers at 7k iterations on OB3D

Six candidate supervision terms. Two landed, two were refuted by measurement, and one refutation was of an idea introduced in this work.

The problem

3DGRUT reconstructs appearance well and geometry poorly. The rendered depth and normal buffers are byproducts of a purely photometric objective, so nothing constrains them to agree with the surface.

Setup for everything that follows

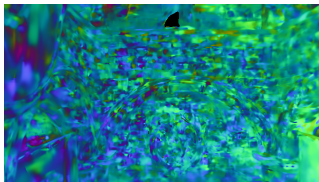
- Three OB3D scenes with ground-truth depth and normals: `sponza`, `lone-monk`, `emerald-square`.
- 7k iterations, gaussian primitive. Reference geometry is used *only* for evaluation; no training term reads it.
- Seed noise, measured over `seed_initialization` 1–4: ± 0.04 – 0.13 dB PSNR, ± 0.002 – 0.004 `abs_rel`, ± 0.1 – 1.2° normal. Effects of a few tenths of a dB need repeated seeds.

The normal metric needs a control. A normal buffer is scored against the angular error of pointing every normal straight back along the view ray — a “normal” that uses no geometry whatsoever. We report `n_gain` = control – error. **Negative `n_gain` means the buffer is worse than using no geometry at all.**

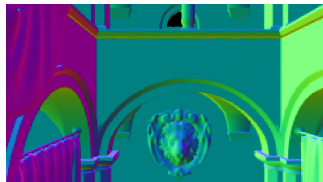
Why the control matters: the baseline normal buffer is noise



rendered RGB — appearance is fine (36.3 dB)



rendered normal — high-frequency noise



reference normal

Sponza, held-out view. The normal buffer is *plausible at a glance* — it has structure and colour variation — and it scores 52.4° mean error with $n_{\text{gain}} = -9.6^\circ$: worse than the geometry-free control.

This is what the whole effort is trying to fix, and it is why the raw angle is not reported alone. A term can improve the angle and still be losing to the control.

Methodology: the scenes fail differently, so scene averages lie

Scene	floaters px	delta1 failures	Failure mode
sponza	0.2–2%	moderate	floaters; per-ray spread detects them (AUC 0.68–0.90)
emerald-square	up to 13%	high	floaters, large depth range
lone-monk	0.01%	17.7%	confidently placed, wrong — no floaters to find

`lone-monk` has essentially no floaters yet the worst `delta1` rate. Its depth is wrong in an *opaque* way: the rays are already concentrated, just at the wrong distance.

Consequence, and a recurring theme of this report: any term that works by concentrating a ray (“commit to one surface”) is structurally unable to help `lone-monk`. A scene average would have hidden that and reported a mild win.

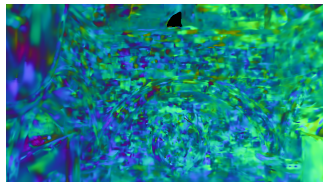
What was attempted

Term	Outcome	One-line verdict
Depth-normal consistency	landed	Transforms normals; best depth on sponza
Depth variance along the ray	refuted	Buys confidence, not accuracy
Scale-z regularisation (PGSR port)	landed, small	Only after fixing a wrong assumption
Pseudo-depth <i>regression</i>	abandoned	Prior is worse than the model it would teach
Pseudo-depth <i>ordinal</i>	landed	Only term improving depth on all 3 scenes
+ its confidence gate	refuted	My own idea; measurement killed it
Multi-view consistency, mesh export	not started	

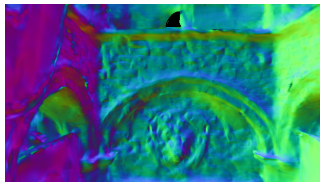
Two of the five conclusions reverse a prediction made before measuring. Those are the slides worth reading.

Depth-normal consistency: tie the normal to the depth gradient

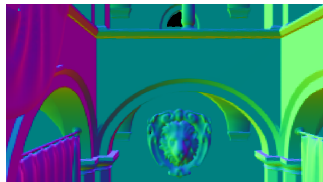
Penalise disagreement between the rendered normal and the normal implied by the rendered depth's local geometry. Purely *internal* — no external data.



baseline: 52.4° , n_gain -9.6



dn 0.05: 25.4° , n_gain $+17.4$



reference

The arch, the masonry courses and the boss are all recovered from a buffer that was noise. Depth improves too: abs_rel -17.0% on sponza.

Cost: -0.2 dB PSNR. And note the scene: on lone-monk the same term gives -0.5% depth — nothing. Section on compounding returns to this.

Depth variance: a sharpening prior with no reference for *where*

Penalising per-ray depth variance asks each ray to commit to a single surface. It looks like a geometry term and behaves like a confidence term.

Why it cannot work as posed

- Variance is zero for a Dirac at *any* distance, so the objective says commit, never where.
- In pairwise form, $M_2 - D^2/\text{acc} = \frac{1}{2_{\text{acc}}} \sum_{ij} w_i w_j (t_i - t_j)^2$, the two-hit case is a **double well** in the near opacity: $w_{\text{near}} w_{\text{far}}$ carries an $\alpha(1 - \alpha)$ factor peaking at $\frac{1}{2}$.
- The near well is **absorbing**: once the front particle reaches $\alpha = 1$, transmittance zeroes the gradient to everything behind it, for every loss. The error is permanent.

Measured consequence

	change
per-ray spread	$\times 0.17$ ($6\times$ tighter)
wrong tight	$\times 170$
sponza abs_rel	$+6\%$
floater fraction	$0.30\% \rightarrow 1.62\%$

It did exactly what it was asked — rays six times tighter — and became 170 times more likely to be confidently wrong.

The same term, rescaled: minima unchanged, behaviour changed

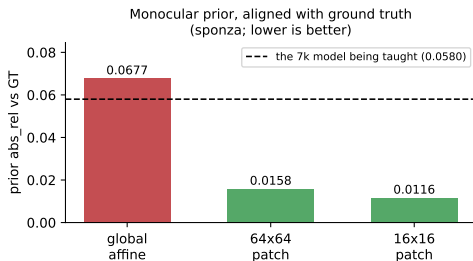
Dividing by μ^2 leaves *both* wells at exactly zero. That was first dismissed as a non-fix — same minima, same problem.

It moved the **separatrix** from 0.49 to 0.82, shrinking the bad basin from 51% of the axis to 18%. Descent responds to the barrier, not to the location of the optima. sponza goes $+6\%$ $\rightarrow -7.8\%$.

Two lessons that outlived the term

- “The minima are unchanged” is not an argument that behaviour is unchanged. When judging a reparameterisation of a degenerate objective, locate the separatrix.
- $\text{Var}/\mu^2 = \text{acc} \cdot M_2/D^2 - 1$ needed *no* new buffer, kernel or backward pass, while the mip-NeRF/2DGS kernel it replaced would have needed a new accumulator on three separate backward paths to reach a strictly worse place. Check what the existing buffers already express before extending the renderer.

The plan called for a scale-invariant regression loss. Measurement said no.



DepthAnythingV2-Base on sponza, affine-aligned against ground truth — the *most favourable* alignment possible, since it uses the ground truth the loss would not have.

- One affine per frame: **0.068**, worse than the **0.058 model it would be teaching**. An aligned L1 would have spent a foundation-model dependency to make depth worse.
- Per 16×16 patch: **0.0116**. The local structure is excellent; the global structure drifts.
- So the signal is not in *any* single affine gauge, and scale-invariance — the exact remedy the plan specified — would not have saved it.

Two further traps: the model emits *disparity* (monotonically inverted, not merely mis-scaled) and *z-depth*, while this renderer's convention is Euclidean ray distance (20% divergence at the corners).

What landed instead: read only the ordering

For a pair of pixels, ask whether the render agrees with the prior about *which is nearer*, and penalise the rendered gap only where they disagree:

$$\mathcal{L} = \frac{1}{|\mathcal{P}|} \sum_{(a,b) \in \mathcal{P}} \text{relu}(- (d_a - d_b) \cdot \text{sign}(-(\text{disp}_a - \text{disp}_b))) / \text{scene extent}$$

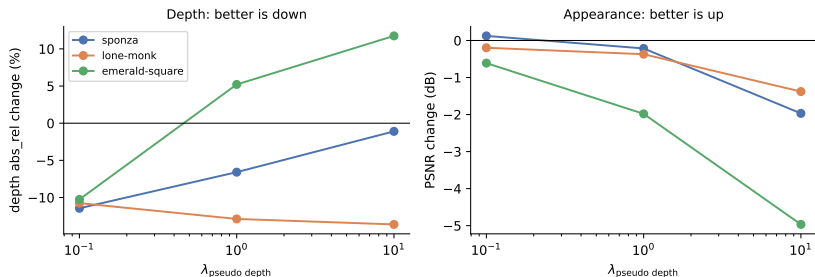
- **Invariant to any increasing transform** of the prior, not just an affine one. The alignment problem, the disparity inversion (one sign flip) and the z-vs-ray-distance mismatch all disappear at once.
- **One-sided**: a pair the render already orders correctly contributes exactly zero *and no gradient*, so the term cannot fight geometry it agrees with.
- Pairs are formed by *cropping* at a random offset $\leq 5\%$ of the image side, not by `torch.roll` — wrapping pairs opposite edges and manufactures disagreements the prior never claimed.

On the pixels where the trained model fails `delta1`, the prior is closer to ground truth **91%** of the time — the population this targets is real.

Result: the first term to improve depth on all three scenes

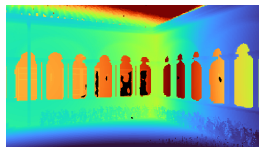
abs_rel, mean $\pm$ sd over 3 seeds	sponza	lone-monk	emerald-square
baseline	0.0571 ± 0.0009	0.0982 ± 0.0019	0.1368 ± 0.0010
ordinal, $\lambda = 0.1$	0.0506 ± 0.0008	0.0876 ± 0.0010	0.1228 ± 0.0033
	-11.4%	-10.8%	-10.3%
PSNR change	+0.12 dB	-0.20 dB	-0.61 dB

A consistent 10–11% at $4\text{--}8\sigma$. Every earlier term had to be argued scene by scene.

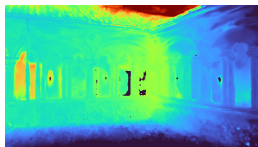


The usable band is narrow: $\lambda = 1$ buys more depth on lone-monk and costs emerald 2.4 dB; $\lambda = 10$ gives up the gain entirely.

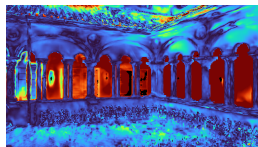
Qualitatively: what the 10% is, and what it is not



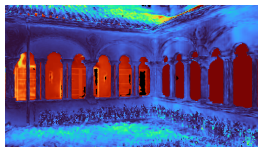
reference depth



baseline



baseline error



ordinal error

lone-monk. Reference depth shows the arch openings as *far* (orange); the baseline puts a confident surface across them at wall depth (green). That is the “confidently placed, wrong” failure no ray-concentration term can reach.

Honest reading of the error maps: the arch openings stay **saturated** — the ordinal term does *not* fix them. What improves is the diffuse error everywhere else: the roof line, the walls and the ground all go blue.

That is the mechanism working as measured, not a disappointment. Pairs are drawn at $\leq 5\%$ offsets and the prior is reliable *locally* (0.0116 per patch), so the term corrects regional depth. A hole in a wall is a long-range discontinuity — exactly the regime where the prior drifts.

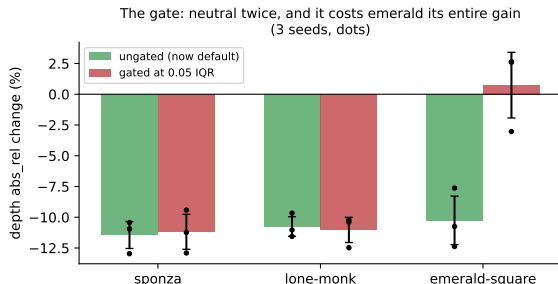
The gate: an idea of mine that measurement killed

The reasoning, before training

Near-tied pairs are where the prior's sign is noise. Drop pairs whose disparity gap is below 0.05 of the map's IQR:

- ordinal agreement with GT: 84% → **97%**
- so ungated, one pair in six pushes the wrong way

This looked decisive. It is wrong.

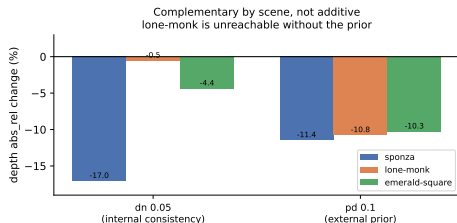


Why it fails, both visible in numbers already collected:

- A large disparity gap selects a *long-range* comparison — precisely where this prior drifts (0.068 global vs 0.0116 per patch). The gate keeps the prior's weakest structure and discards its best.
- For a *one-sided* loss, 97% agreement is a warning, not a benefit: only 3% of surviving pairs can produce any gradient at all.

The offline metric counted pairs instead of asking what they teach. Default is now off, pinned by a test so re-enabling it has to be re-measured.

The terms fix different scenes; they do not add



Internal consistency vs external prior

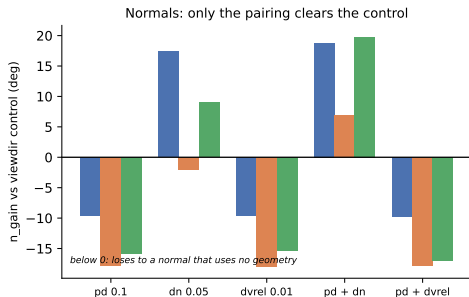
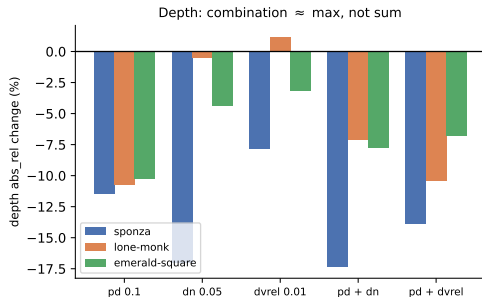
- Depth-normal consistency owns sponza (-17.0%) and is null on lone-monk (-0.5%).
- The ordinal prior is the reverse: even on lone-monk it gives -10.8% .

This is the scene-split slide made concrete.

lone-monk's error is not an inconsistency, so no internal condition detects it; it takes information from outside the scene. **Combined, depth is**

sub-additive — the pair lands near the *max* of the two, never the sum, and on lone-monk it is **worse than the prior alone** (-7.1% vs -10.8%).

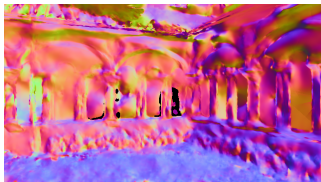
But on normals the pairing is where everything pays off



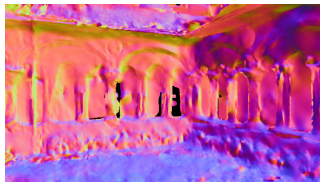
The ordinal term **alone does nothing for normals** ($52.4^\circ \rightarrow 52.5^\circ$, still below the control) — expected, since the rendered normal is the particle's local z axis and depth ordering does not reach it.

Yet $pd + dn$ is the only configuration whose normals clear the control on all three scenes. Depth-normal consistency ties the normal to the depth *gradient*, so it can only be as good as the depth handed to it: the prior improves the depth, and consistency carries that across into normals.

The cross-channel effect, seen directly



dn alone: 37.4° , n_gain -2.1



pd + dn: 28.3° , n_gain $+6.9$



reference

lone-monk normals. With depth-normal consistency alone the columns and arches are blobby and the buffer **still loses to the geometry-free control** (n_gain -2.1°). Adding a depth term that touches no normals resolves the colonnade and flips the sign ($+6.9^\circ$).

The lesson generalises: before concluding a depth term is irrelevant to normals, check the cross-channel pairing.

A methodological trap worth one slide

The pairing of the ordinal term with relative depth variance was first measured at $\lambda = 1$ for **both** — $10\times$ and $100\times$ their measured optima — and written up as a null.

depth abs_rel change	sponza	lone-monk	emerald-square
ordinal alone ($\lambda = 0.1$)	-11.4%	-10.8%	- 10.3%
variance alone ($\lambda = 0.01$)	-7.9%	+1.2%	-3.2%
both, at their own optima	- 13.9%	-10.4%	-6.8%
both at $\lambda = 1$ (the first pass)	-7.6%	-12.9%	+10.3%

At sane weights the verdict changes from *harmful* to *scene-dependent*: the pair beats both single terms on `sponza`, is a wash on `lone-monk`, and is still clearly **worse than the prior alone** on `emerald-square`. That is a weaker claim than the first pass suggested in the other direction, and it is the one the data supports.

A null from a combination measured at the wrong weights is not a null about the combination. The over-weighted cell is kept in the sweep so the trap stays visible.

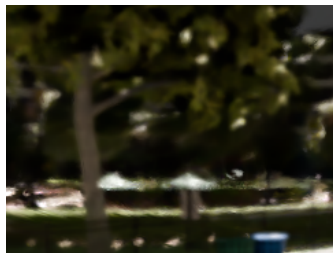
The photometric cost, seen directly: it is a real regression



reference



baseline — 32.9 dB



pd + dn — 32.0 dB

emerald-square foliage, the scene and the region carrying the largest cost. Two things happen at once:

- The baseline's **spiky starburst artifacts** (left of frame, mid-left) are **gone** — they are not in the reference either.
- The genuine foliage structure is **smoothed away** with them.

Against the reference the blur costs more than the artifacts did: mean absolute error 4.28 $\rightarrow$ 4.80, with $pd + dn$ closer to the reference on only 41% of pixels. Vegetation is where a consistency term assuming a locally planar surface should be expected to pay most, and it does.

Recommended settings

Goal	Configuration	Cost
Depth, one term, no tuning	pd 0.1 alone –10 to –11% on all three scenes; the safe default	0–0.6 dB
Depth <i>and</i> normals	pd 0.1 + dn 0.05 the only config clearing the normal control on all 3 scenes, and the best sponza depth measured (–17.3%)	0.2–0.9 dB

pd 0.1 + dvrel 0.01 is the best *depth-only* pairing on sponza (–13.9%) at a smaller photometric cost (0.1–0.2 dB), but it loses to pd alone on emerald-square, so it is not a general recommendation.

All of these remain **off by default**. The photometric cost is real, and PSNR is what the codebase is currently optimised and evaluated for.

Limitations, stated plainly

- **Everything is at 7k iterations.** No term has been confirmed at 30k. Gates tuned for a 7k sweep (`*_from_iter`) do not transfer unchanged.
- **Three scenes.** They were chosen because they fail differently, not because they are representative.
- **Combination cells are single-seed;** single-term cells are 3-seed. The `lone-monk` normal sign flip and the `sponza` depth gains are far outside seed noise; `emerald` differences of ~ 1 point are not.
- **The emerald-square PSNR cost (0.6–0.9 dB) is only half explained.** It is spatially concentrated in foliage, which is measured. *Why* that scene suffers most — largest depth range, so the prior’s global drift should hurt most — remains a hypothesis.
- **No mesh.** Nothing here has been evaluated by surface extraction or Chamfer distance, which is what “better geometry” would ultimately mean.

What the exercise actually taught

1. **Always include a control that uses no information.** The baseline normal buffer looked reasonable and was worse than pointing every normal at the camera. Half the terms here would have been “improvements” without `n_gain`.
2. **Diagnose the scene, not the average.** `lone-monk` has no floaters and the worst depth; a whole family of terms is structurally incapable of helping it, and a scene mean would have reported a win.
3. **An offline metric that motivates a mechanism can be measuring the wrong thing.** The gate’s 84%→97% was real and irrelevant: it counted pairs instead of asking what they teach.
4. **Check what the reference quantity means *here*.** “Monocular depth” is not depth up to one affine; PGSR’s “min scale” is not this renderer’s normal axis. Both cost a regression before being caught.
5. **Measure combinations at each term’s own optimum**, and against each single term — not only against the baseline.

Summary

Two terms landed. Ordinal supervision from a monocular prior improves depth 10–11% on all three scenes — the only term here to help all of them — and, paired with depth-normal consistency, produces the only normal buffer that beats a geometry-free control everywhere.

Three ideas were refuted by measurement, one of them introduced in this work. The refutations are recorded in `docs/normal-supervision.md` alongside the predictions they contradict, because the reasoning that failed is the part that generalises.